

TECHNICAL FINDINGS & PROJECT SUMMARY

Groundwater & GIS Portfolio

Two paired QGIS + MODFLOW 6/FloPy projects
— Metro Vancouver regional screening and the
New Afton (Kamloops) mine site — documenting
methods, results, and what each demonstrates.

Prepared by Mogli · Vancouver, BC

BSc Environmental Science

QGIS 3.44 (Solothurn) · MODFLOW 6.7.0 / FloPy

14-day field methods course — soil/water sampling, equipment
calibration, chain-of-custody

This document summarizes two self-directed technical projects built to develop skills used in junior environmental technician, GIS analyst, and hydrogeology technician roles. Each site pairs a QGIS layer — mapping the real physical controls on groundwater (contamination, terrain, structure) — with a numerical MODFLOW 6 flow model built in FloPy that reflects those same controls. The findings below are the actual model output, re-verified by re-running both models directly from source rather than summarized from memory.

WHY THIS IS WORTH A CLOSER LOOK

Most entry-level environmental applicants can list "QGIS" and "Python" on a resume. Far fewer have carried a project through the full pipeline this repo shows twice: map the physical controls (contamination plume, terrain, fault structure) → translate that into a numerical model with matching boundary conditions → run it to convergence → interpret the result → export it back into the GIS project it came from. That loop — not any single tool — is the actual day-to-day of a junior hydrogeology or GIS technician role, and it's demonstrated here twice, at two different scales (a 40 km regional screen and a site-scale mine model), with the limitations of each stated plainly rather than glossed over.

Site 1 — Metro Vancouver

Regional groundwater vulnerability screening · QGIS 3.44 · EPSG:3005 (BC Albers)

OBJECTIVE

Build a regional GIS base to screen where groundwater in Metro Vancouver is most exposed to contamination and terrain risk, then pair it with a numerical flow model over the same domain to show how the water table responds to a pumping stress.

GIS LAYER — WHAT WAS BUILT

The project layers span administrative boundaries (GADM, 4 levels), a DEM clipped to the study extent, a provincial groundwater aquifer classification layer, GWELLS well points, and a Ramsar wetlands layer for ecological context. Two layers were derived rather than sourced directly: point petroleum hydrocarbon (PHC) concentrations were interpolated into a continuous plume surface using QGIS's IDW interpolation tool, and the DEM was reclassified with the raster calculator to isolate terrain over 15° slope, flagging ground where runoff and erosion would complicate a vulnerability read. Both derived layers were cross-referenced against the aquifer and well layers, then clipped and exported through Print Layout.

LAYER	TYPE	ROLE
PHC_IDW_PLUME	derived raster	interpolated contamination surface
Rasterslope>15	derived raster	terrain risk screen
Ground Water Aquifers	vector	provincial aquifer classification
well	vector	GWELLS well points
Ramsar sites	vector	ecological reference

FLOW MODEL — WHAT WAS BUILT

A single-layer, steady-state MODFLOW 6 model was built over the same domain to test the water table's response to a central pumping well between two fixed-head boundaries.

PACKAGE	PURPOSE	SETTING
DIS	grid	40 × 40 cells, 100 m spacing, top 80 m / bottom 50 m
NPF	conductivity	K = 15 m/day, uniform
RCHA	recharge	0.0009 m/day
CHD	boundaries	75 m (west) / 65 m (east)
WEL	pumping	-300 m³/day, domain centre

RESULT

Heads ranged 65.0–76.3 m across the domain, declining smoothly from the inflow to outflow boundary with a visible local drawdown at the pumping well.

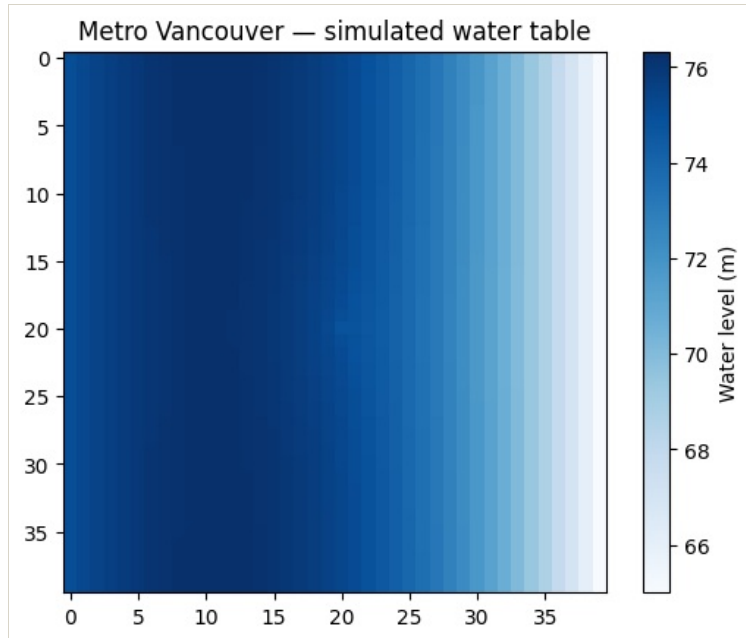


Fig. 1.1 — raw model output, unedited.

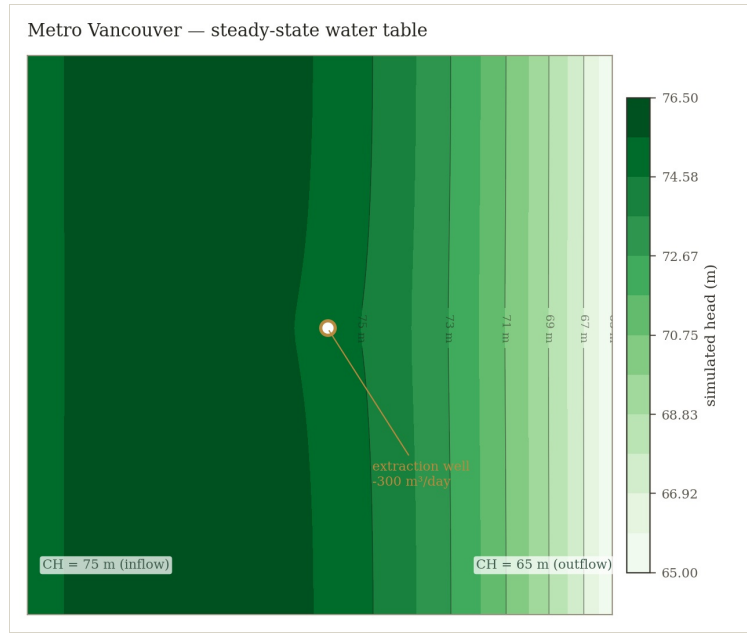


Fig. 1.2 — same run, contoured with boundary and well labels.

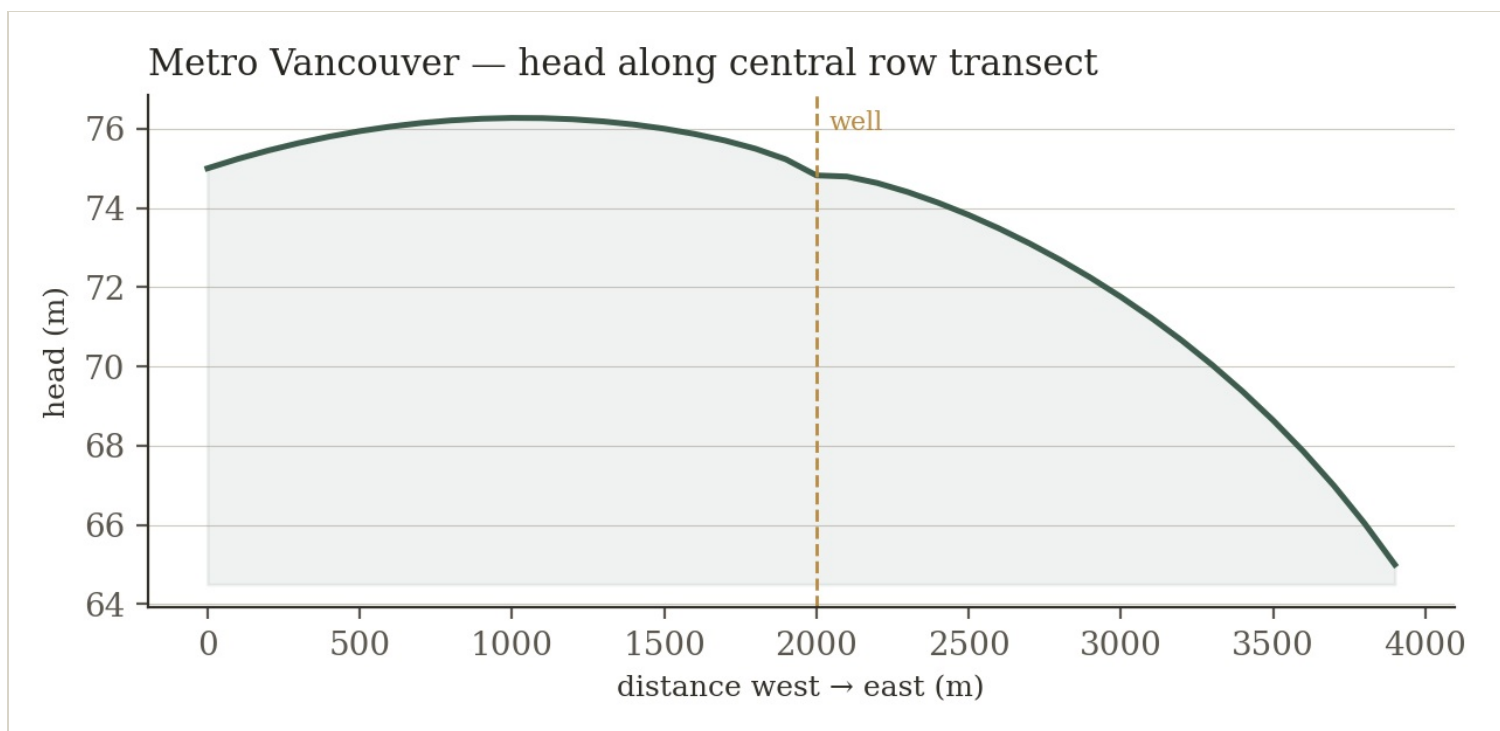


Fig. 1.3 — head along the central row: a small recharge mound west of the well, then decline to the outflow boundary.

Honest limitation — K and recharge are literature-typical values, not field-calibrated. This proves out the build→run→visualize→export pipeline correctly; the credible next step is checking the model against 2–3 observed heads from the GWELLS layer.

Site 2 — Kamloops / New Afton Mine Site

Site-scale monitoring network & structural flow model · QGIS 3.44

OBJECTIVE

Map the structural controls on groundwater flow at the mine site, then use them to justify a defensible monitoring well network and a flow model that reflects the same fault-zone geometry — the kind of paired deliverable used in mine-site environmental monitoring programs.

GIS LAYER — MONITORING NETWORK DESIGN

Layers include the DEM, mine footprint (*Newaftonmine*) and city extent (*KamloopsCity_Area*), mapped structural features (*Hangingwall_fault_points*, *Oblique_slip_faultpts*), and existing well data. On top of this, an 8-point monitoring network was placed by hydrogeological function rather than even spacing:

WELL (S)	FUNCTION
SW-01	Background — upgradient baseline, west of the mine
SW-02, SW-03	Source — near-pit source zone
SW-04, SW-05	Interceptor — early warning, mid-pathway
SW-06, SW-07	Sentinel — protecting the Kamloops supply corridor
SW-08	Compliance — trigger point nearest Kamloops

FLOW MODEL — WHAT WAS BUILT

A single-layer model with a two-zone hydraulic conductivity field, reflecting the mapped fault corridor as a high-K band against a low-K background, stressed by two extraction wells.

PACKAGE	PURPOSE	SETTING
DIS	grid	30 × 30 cells, 20 m spacing, top 380 m / bottom 340 m
NPF	conductivity	zoned — 0.5 m/day background, 25 m/day fault-corridor band
RCHA	recharge	0.00004 m/day
CHD	boundaries	375 m (south) / 350 m (north)
WEL	pumping	two wells, -800 m³/day each, low-K zone

RESULT

Heads spanned 350–375 m, matching the boundary conditions exactly, with the gradient steepening through the low-K zone and flattening sharply once it crosses into the high-K corridor.

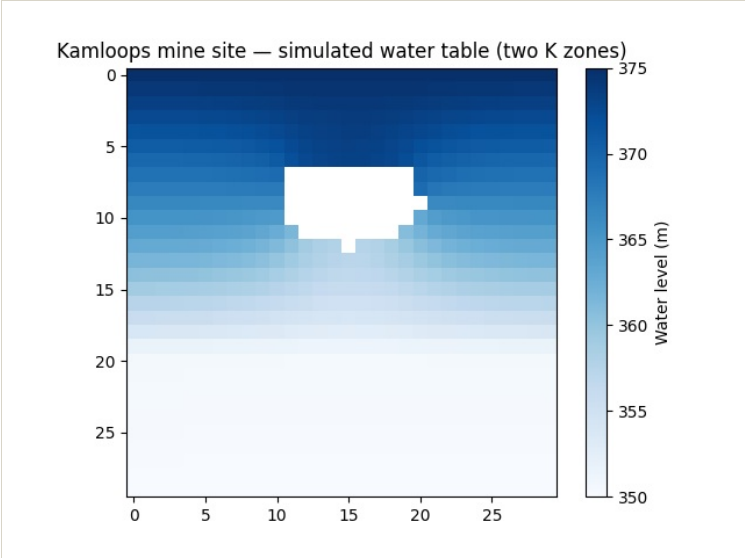


Fig. 2.1 — raw model output, same plotting code as the notebook.

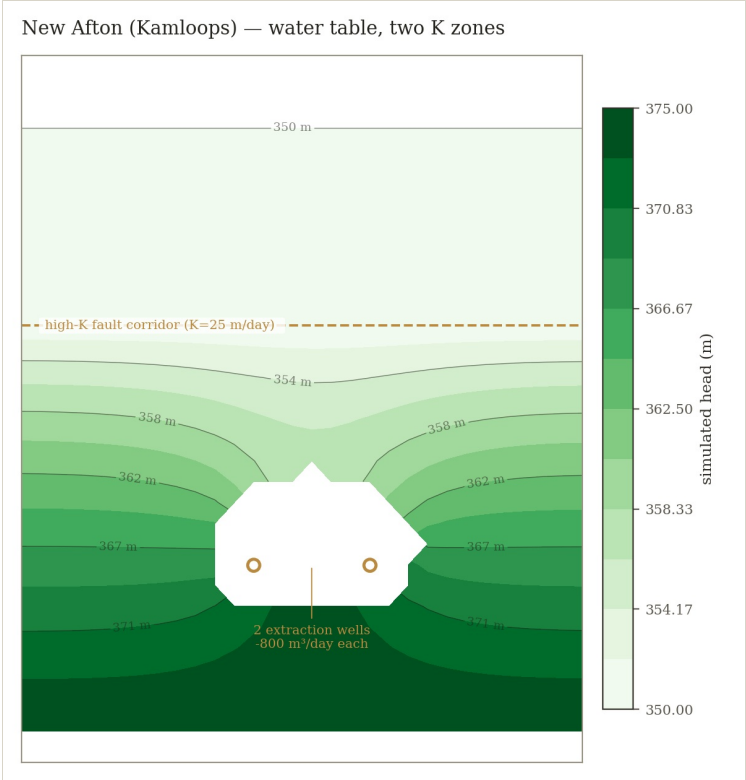


Fig. 2.2 — same run, contoured with the fault corridor and wells labelled.

Kamloops — head along west-edge column, low-K → high-K

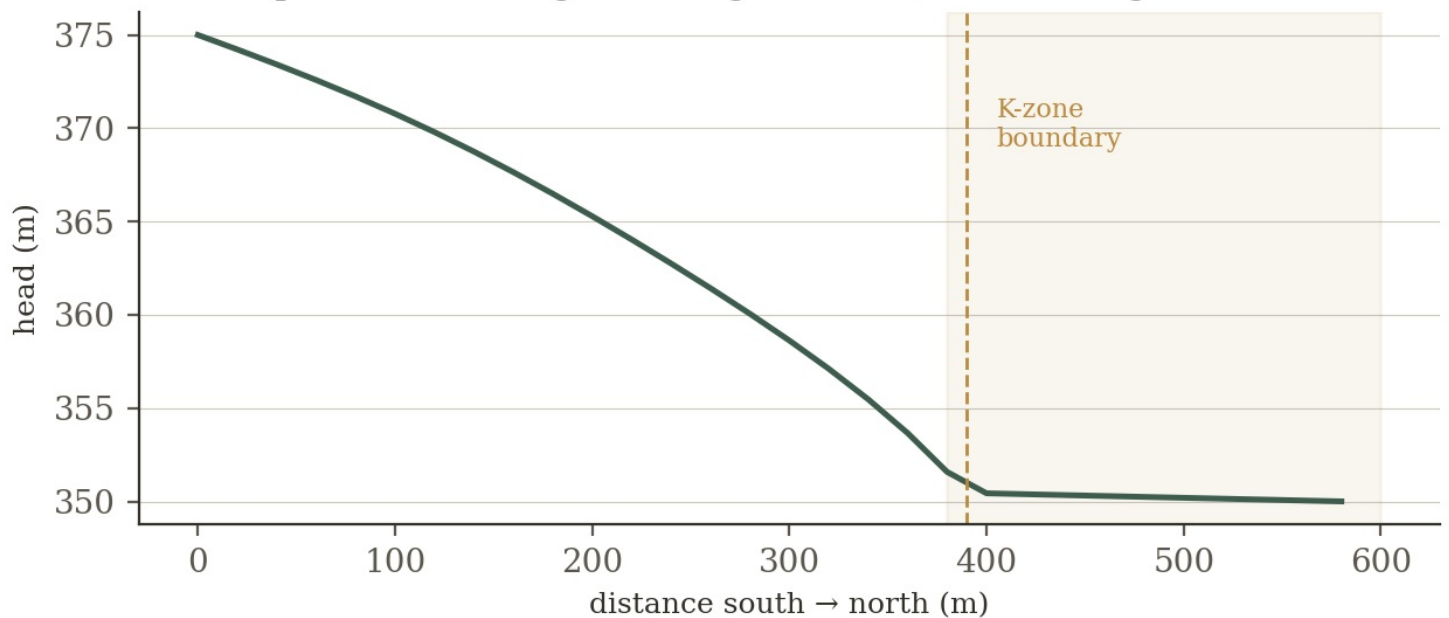


Fig. 2.3 — head along a column clear of the wells: steep gradient in the low-K background, flattening sharply in the high-K corridor — the textbook signature of a preferential flow path, and the clearest evidence that the K-zonation is doing something physically real.

Honest limitation — the K-zone boundary was hand-set from the mapped fault line rather than extracted programmatically from the fault point coordinates, and — depending on solver version — a few cells directly around the wells can come back dry, since $-800 \text{ m}^3/\text{day}$ is a lot to ask of a 0.5 m/day zone at this grid resolution. That's a real finding about the well design (split the rate, or move wells closer to the corridor), not an error.

03 HOW THE TWO HALVES CONNECT, AT EACH SITE

GIS → MODEL

At both sites, the model's boundaries, well placement, and (at Kamloops) hydraulic zonation were set to match what's mapped in QGIS — the fault corridor, the mine footprint, the plume extent — chosen by eye from the GIS layers rather than extracted programmatically.

MODEL → GIS

Output head arrays export as GeoTIFFs in the same coordinate system (EPSG:3005) as the source QGIS projects, so results can be brought back in and displayed against the plume and fault layers they were built from.

WHAT WOULD CLOSE THE LOOP

Pulling fault point coordinates directly from the .qgz file to set K-zone geometry programmatically, and importing well coordinates as calibration targets instead of hand-typed values.